

Laboratoire 4a

243-436 – Mesures EMC – Émissions Conduites



Cégep Limoilou

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1 But

Prendre les mesures des émissions conduites d'un circuit électronique.

2 Matériels

- PC avec Logiciel EMCvu
- Analyseur de spectre Tektronix RSA503, câble d'alimentation et câble USB.
- LISN TBOH01 Tekbox
- Jeux de câble BNC et type-N

- E.S.T. : Circuit '**Alimentation à découpage**', réalisé dans le cours d'électronique
- Charge de test

3 Préparation

Explication et démonstration par le professeur.

3.1 Diagramme de l'E.S.T. (Mesures d'émissions conduites)



3.2 Mode de Fonctionnement

...

1. Charge à vide
2. Pleine Charge (résistive)
3. Charge – exemple Moteur (Puissance ?)

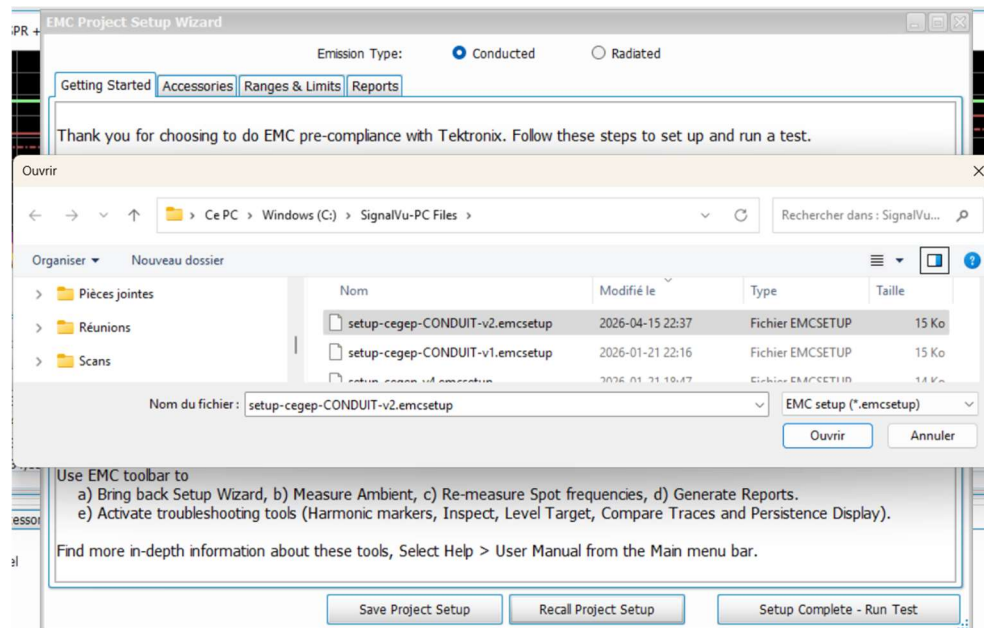
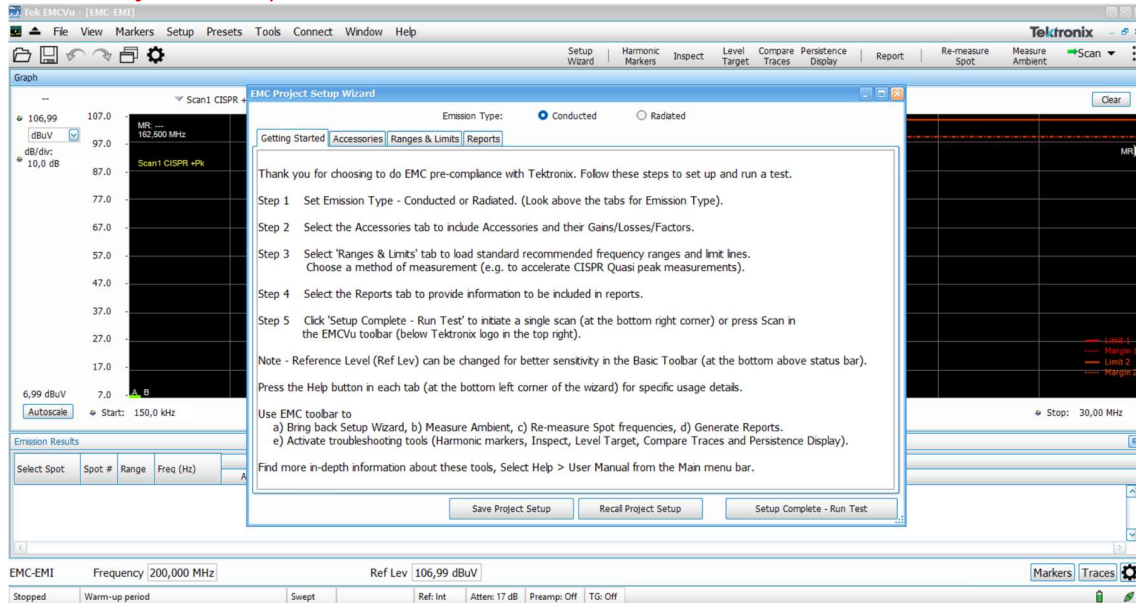
...

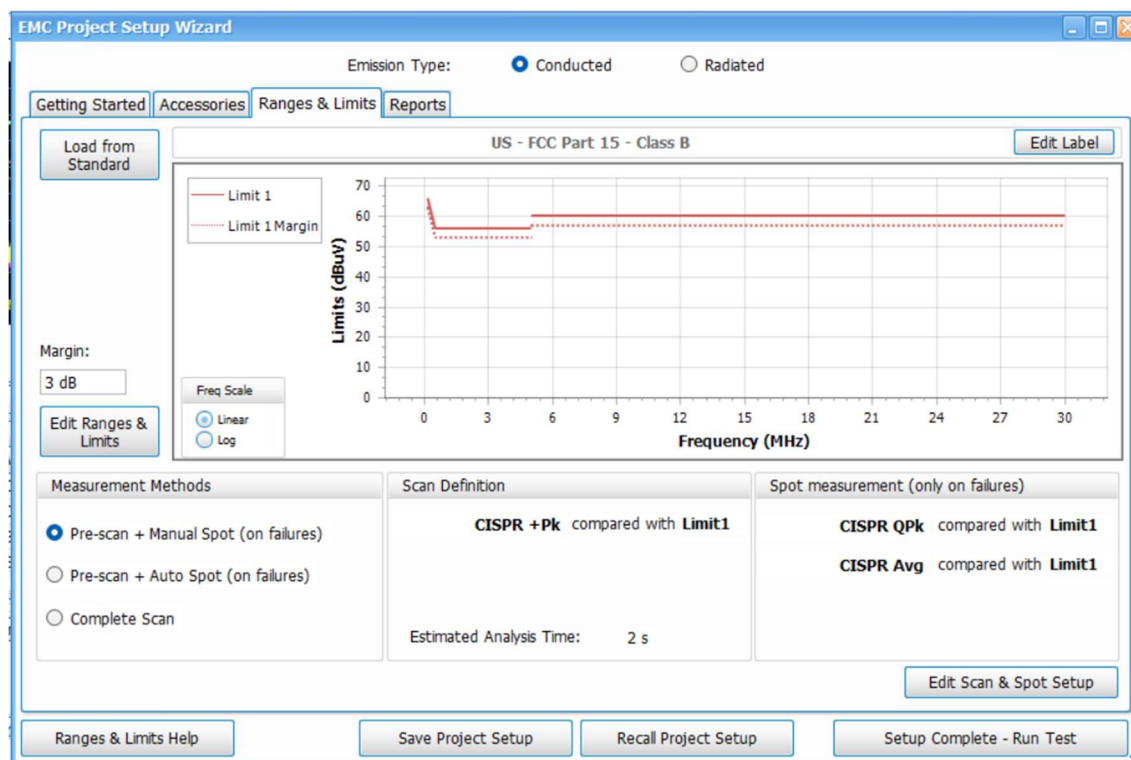
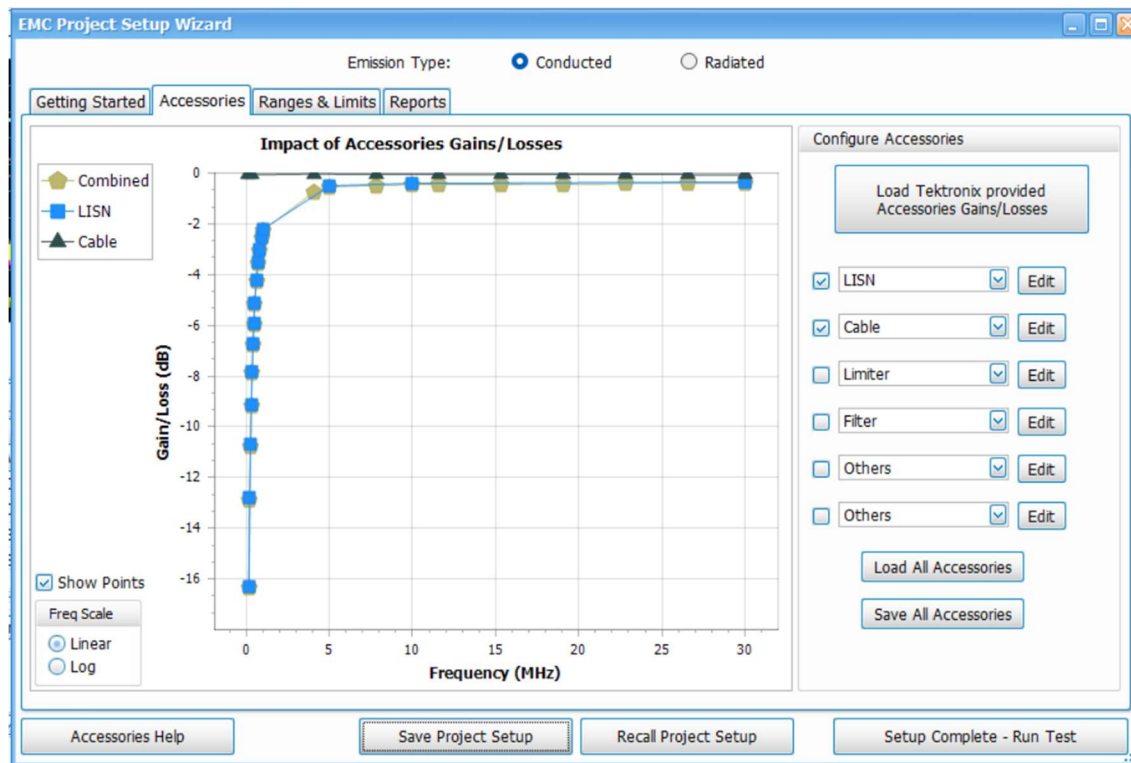
3.3 Consignes

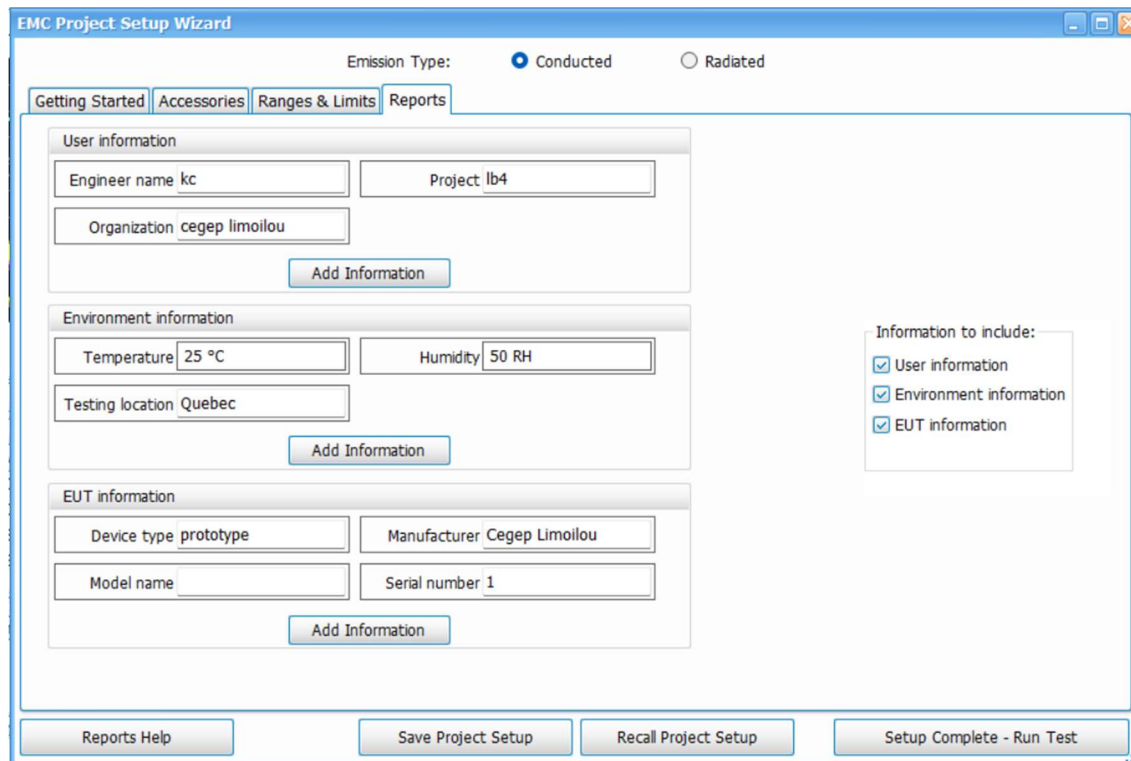
- Préparer le setup de test
- Configurer le logiciel
 - Emission Type : Conducted
 - Accessories :
 - LISN
 - Cable
 - Ranges and limits :
 - FCC Part 15 class B
 - Reports
 - Name
 - Projet
 - Device type
 - Manufacturer
- Prendre la mesure du bruit de fond (Measure Ambient)
- Prendre les mesures des différents modes de test (Scan)
 - 1^{er} mode de fonctionnement
 - Sauvegarder les résultats pour le rapport (report)
 - 2^{ème} mode de fonctionnement
 - Sauvegarder les résultats pour le rapport
 - 3^{ème} mode de fonctionnement
 - Sauvegarder les résultats pour le rapport
- Générer et sauvegarder le rapport

4 Analyseur de Spectre et Logiciel EMCvu

4.1 Project Setup Wizard







The image shows a screenshot of the "EMC Project Setup Wizard" software window. The window has a blue title bar and a standard Windows-style interface. At the top, there are tabs for "Getting Started", "Accessories", "Ranges & Limits", and "Reports". The "Getting Started" tab is active. Below the tabs, there is a section for "Emission Type:" with two radio buttons: "Conducted" (selected) and "Radiated".

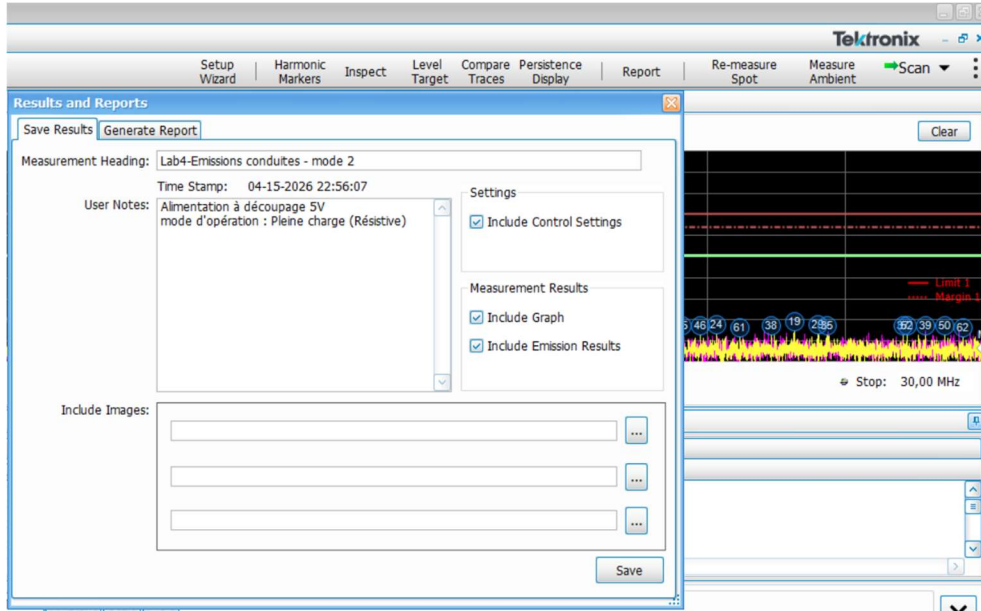
The main area of the wizard is divided into three sections, each with a "Add Information" button:

- User information:** Contains fields for "Engineer name" (filled with "kc"), "Project" (filled with "lb4"), and "Organization" (filled with "cegep limoilou").
- Environment information:** Contains fields for "Temperature" (filled with "25 °C"), "Humidity" (filled with "50 RH"), and "Testing location" (filled with "Quebec").
- EUT information:** Contains fields for "Device type" (filled with "prototype"), "Manufacturer" (filled with "Cegep Limoilou"), "Model name" (empty), and "Serial number" (filled with "1").

On the right side of the wizard, there is a section titled "Information to include:" with three checked checkboxes: "User information", "Environment information", and "EUT information".

At the bottom of the window, there are four buttons: "Reports Help", "Save Project Setup", "Recall Project Setup", and "Setup Complete - Run Test".

4.2 Rapport



Results and Reports

Save Results | **Generate Report**

Measurement Heading: Lab4-Emissions conduites - mode 2

Time Stamp: 04-15-2026 22:56:07

User Notes: Alimentation à découpage 5V
mode d'opération : Pleine charge (Résistive)

Settings

- ☒ Include Control Settings

Measurement Results

- ☒ Include Graph
- ☒ Include Emission Results

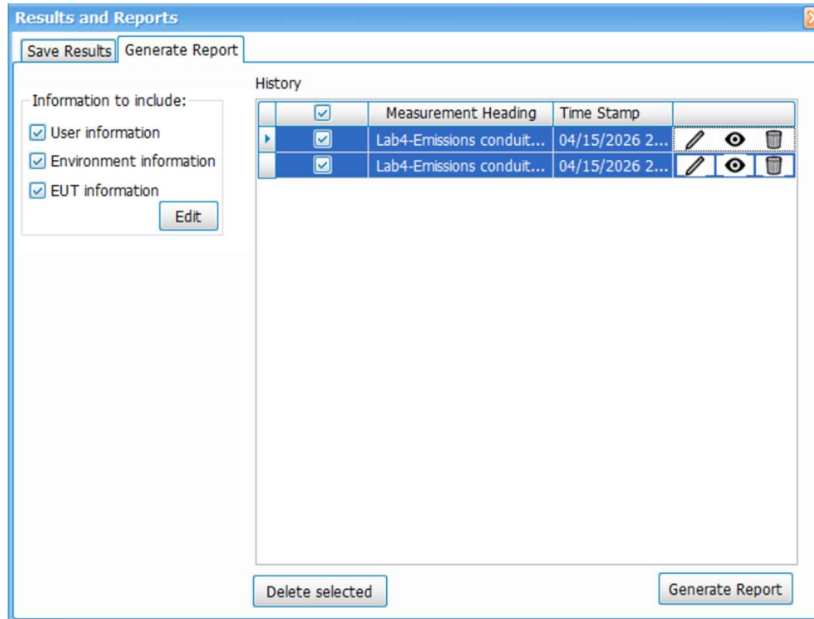
Include Images:

Save

Clear

Limit 1
Margin 1

Stop: 30,00 MHz



Results and Reports

Save Results | **Generate Report**

Information to include:

- ☒ User information
- ☒ Environment information
- ☒ EUT information

Edit

History

		Measurement Heading	Time Stamp			
☒		Lab4-Emissions conduit...	04/15/2026 2...			
☒		Lab4-Emissions conduit...	04/15/2026 2...			

Delete selected

Generate Report

5 Annexe

Fichier de configuration – émissions conduites – Tektronix EMCvu

setup-cegep-CONDUIT-v2.emcsetup

```
<?xml version="1.0" encoding="utf-8"?>
<!--This file is generated by the EMC Vu program.-->
<EMCSetup>
  <EMCLevelTargetValue>63,49</EMCLevelTargetValue>
  <EMCLevelTargetEnabled>True</EMCLevelTargetEnabled>
  <EMCReportSetup>
    <UserReportSettingsEnabled>True</UserReportSettingsEnabled>
    <EnvironmentReportSettingsEnabled>True</EnvironmentReportSettingsEnabled>
    <EUTReportSettingsEnabled>True</EUTReportSettingsEnabled>
    <UserSettings>
      <EMCFieldNameUserSettings1>Engineer name</EMCFieldNameUserSettings1>
      <EMCFieldValueUserSettings1>kc</EMCFieldValueUserSettings1>
      <EMCFieldNameUserSettings2>Project</EMCFieldNameUserSettings2>
      <EMCFieldValueUserSettings2>lb4</EMCFieldValueUserSettings2>
      <EMCFieldNameUserSettings3>Organization</EMCFieldNameUserSettings3>
      <EMCFieldValueUserSettings3>cegep limoilou</EMCFieldValueUserSettings3>
    </UserSettings>
    <EnvironmentSettings>
      <EMCFieldNameEnvironmentSettings1>Temperature</EMCFieldNameEnvironmentSettings1>
      <EMCFieldValueEnvironmentSettings1>25</EMCFieldValueEnvironmentSettings1>
      <EMCFieldNameEnvironmentSettings2>Humidity</EMCFieldNameEnvironmentSettings2>
      <EMCFieldValueEnvironmentSettings2>50</EMCFieldValueEnvironmentSettings2>
      <EMCFieldNameEnvironmentSettings3>Testing location</EMCFieldNameEnvironmentSettings3>
      <EMCFieldValueEnvironmentSettings3>Quebec</EMCFieldValueEnvironmentSettings3>
    </EnvironmentSettings>
    <EUTSettings>
      <EMCFieldNameEUTSettings1>Device type</EMCFieldNameEUTSettings1>
      <EMCFieldValueEUTSettings1>prototype</EMCFieldValueEUTSettings1>
      <EMCFieldNameEUTSettings2>Manufacturer</EMCFieldNameEUTSettings2>
      <EMCFieldValueEUTSettings2>Cegep Limoilou</EMCFieldValueEUTSettings2>
      <EMCFieldNameEUTSettings3>Model name</EMCFieldNameEUTSettings3>
      <EMCFieldValueEUTSettings3 />
      <EMCFieldNameEUTSettings4>Serial number</EMCFieldNameEUTSettings4>
      <EMCFieldValueEUTSettings4>1</EMCFieldValueEUTSettings4>
    </EUTSettings>
  </EMCReportSetup>
  <EMCRangesAndLimitsSetup>
    <Limit2Enabled>False</Limit2Enabled>
    <Limit3Enabled>False</Limit3Enabled>
    <Margin>3</Margin>
    <LimitLabel>US - FCC Part 15 - Class B</LimitLabel>
    <Ranges>
      <Range>
        <Name>A</Name>
        <Enabled>True</Enabled>
        <StartFrequency>150000</StartFrequency>
        <StopFrequency>500000</StopFrequency>
```